THE PRESENTATION CHECKLIST

THIS SLIDE IS A LIST OF RECOMMENDATIONS FOR BEST PRACTICE WHEN PRESENTING. DELETE IT FROM YOUR FINAL PRESENTATION.

Online Presentation Tip

If you are presenting in Zoom, it may be helpful to leave space for the window with your face. An area approximately equivalent to this box might be appropriate. To add it to your slide master go to https://go.unimelb.edu.au/9uc8.

- Have you :
  - Included 7 **elements** or fewer per slide
  - Kept text to 18-32 point **size font** (slightly smaller for online)
  - Kept visuals clean and simple
  - Used professional animations to reveal information in logical **chunks**
  - Highlighted **key ideas** using transparent boxes
  - **Acknowledged** images where needed
  - De-emphasised and/or emphasised data to only **include what is necessary**
  - Kept a strong **contrast** between text and images
  - Used a professional, consistent **colour pallet**
- For more information on the above checklist, refer to the Oral Presentations Module in the LMS

INFO MANAGEMENT 1: DATA QUALITY

- The difference between data, information, and knowledge.
- The dimensions of data quality
- The article to be analysed, the assertions made, and the associated sources.

ARTICLE REQUIREMENTS

- It should relate to an engineering/IT topic.
- It should make assertions about the current or future state of the topic.
- It should support the assertions with external sources.

POST MEETING PLAN…

- Meet again on Tuesday
- Avryl start preparing background slides
- FuQuan find an appropriate article for the activity

DATA QUALITY

Information Management

ENGR90033:Internship – Seminar Presentation

	Presenter 1	Presenter 2
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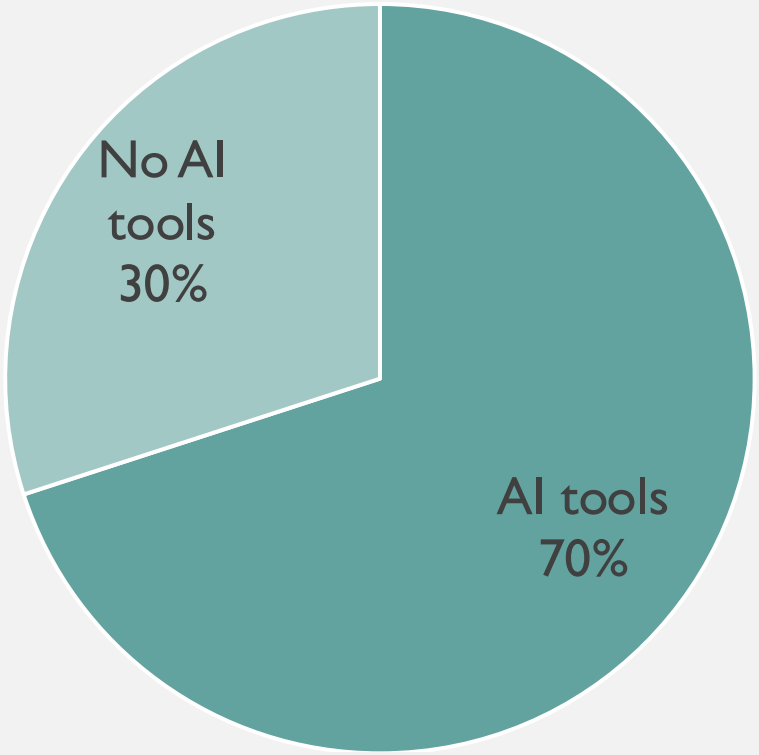
ACTIVITY 1

Intuitive judgment

ACTIVITY 1: INTUITIVE JUDGEMENT

Assertion

- 70% of developers are already using or planning to use AI tools.



Source: Stack Overflow Developer Survey 2023 (cited in Werner Vogels' article)

*https://www.allthingsdistributed.com/2023/11/tech-predictions-for-2024-and-beyond.html
*https://survey.stackoverflow.co/2023/#sentiment-and-usage-ai-sel-prof

ACTIVITY 1

What is your intuitive judgment base on the information given?

(1 = Not at all credible, 10 = Highly credible)

Data

Information

Knowledge

Data

Definition:
Raw, unprocessed facts without context or meaning.

Example:
- Millions of raw logs generated per second by servers
- Individual survey responses (e.g. "Yes, I use AI tools")

In this article:
- Raw responses collected from the Stack Overflow survey
- Each developer's individual answer about AI usage

Information

Knowledge

Data

Information

Definition:
Data that has been processed, organised, and given context.

Example:
"70% of developers are using or planning to use AI tools"

How it is formed:

- Aggregating individual survey responses
- Calculating percentages
- Presenting results in a meaningful way

In this article:
The Stack Overflow survey result becomes a statistical claim

Knowledge

Data

Information

Knowledge

Definition:
The interpretation and application of information to support decisions or predictions.

Example:
- AI tools will become essential in software development
- Developer productivity will be redefined

In this article:
- Werner Vogels uses the survey result to predict that: "AI assistants will redefine developer productivity"

This represents a shift from data to strategic insight.

DIMENSIONS OF DATA QUALITY

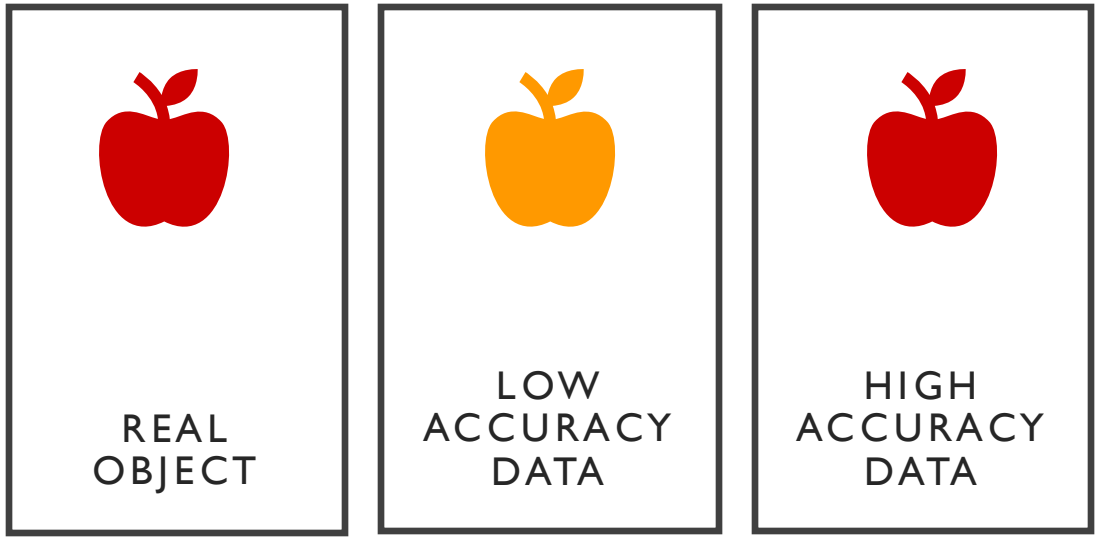
- Accuracy
- Completeness
- Consistency
- Fitness for purpose
- Timeliness
- Uniqueness
- Validity

DIMENSIONS OF DATA QUALITY

Accuracy

- Correctness of data values
- Assessed based on agreed upon "source of truth"

MEASURE OF HOW CLOSELY THE DATA VALUES MATCH THE DATA SOURCE



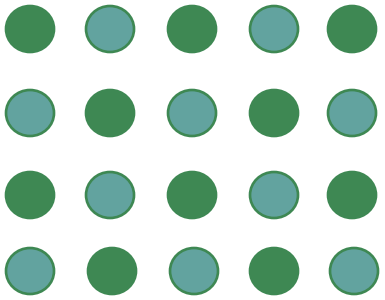
DIMENSIONS OF DATA QUALITY

Completeness

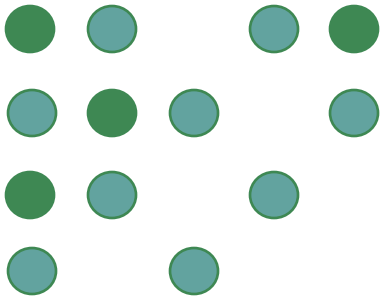
AMOUNT OF DATA THAT IS

- USEABLE
- COMPLETE

COMPLETE DATA (REPRESENTATIVE SAMPLE)



MISSING VALUES (DATA IS NOT REPRESENTATIVE)



DIMENSIONS OF DATA QUALITY

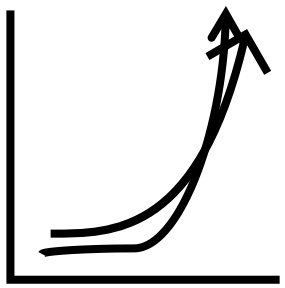
Consistency

- Between data from different sources

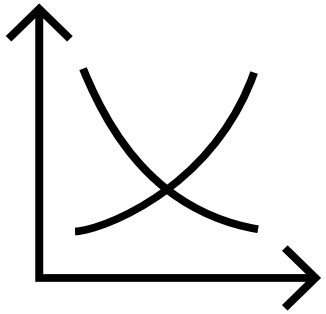
CHECKING FOR

- CONSISTENT DATA TRENDS
- DATA RELATIONSHIPS

CONSISTENT DATA



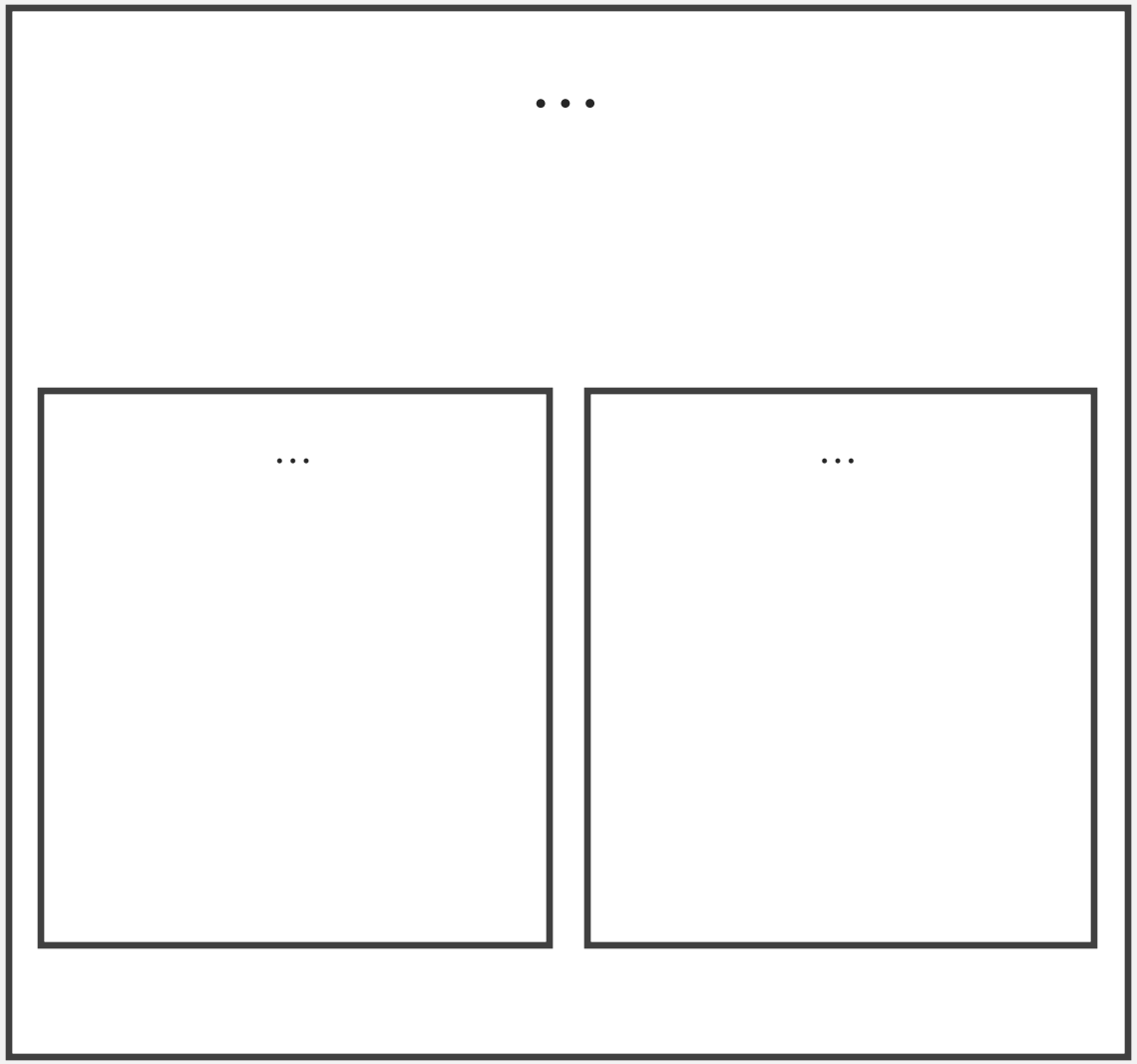
INCONSISTENT DATA



DIMENSIONS OF DATA QUALITY

Fitness for purpose

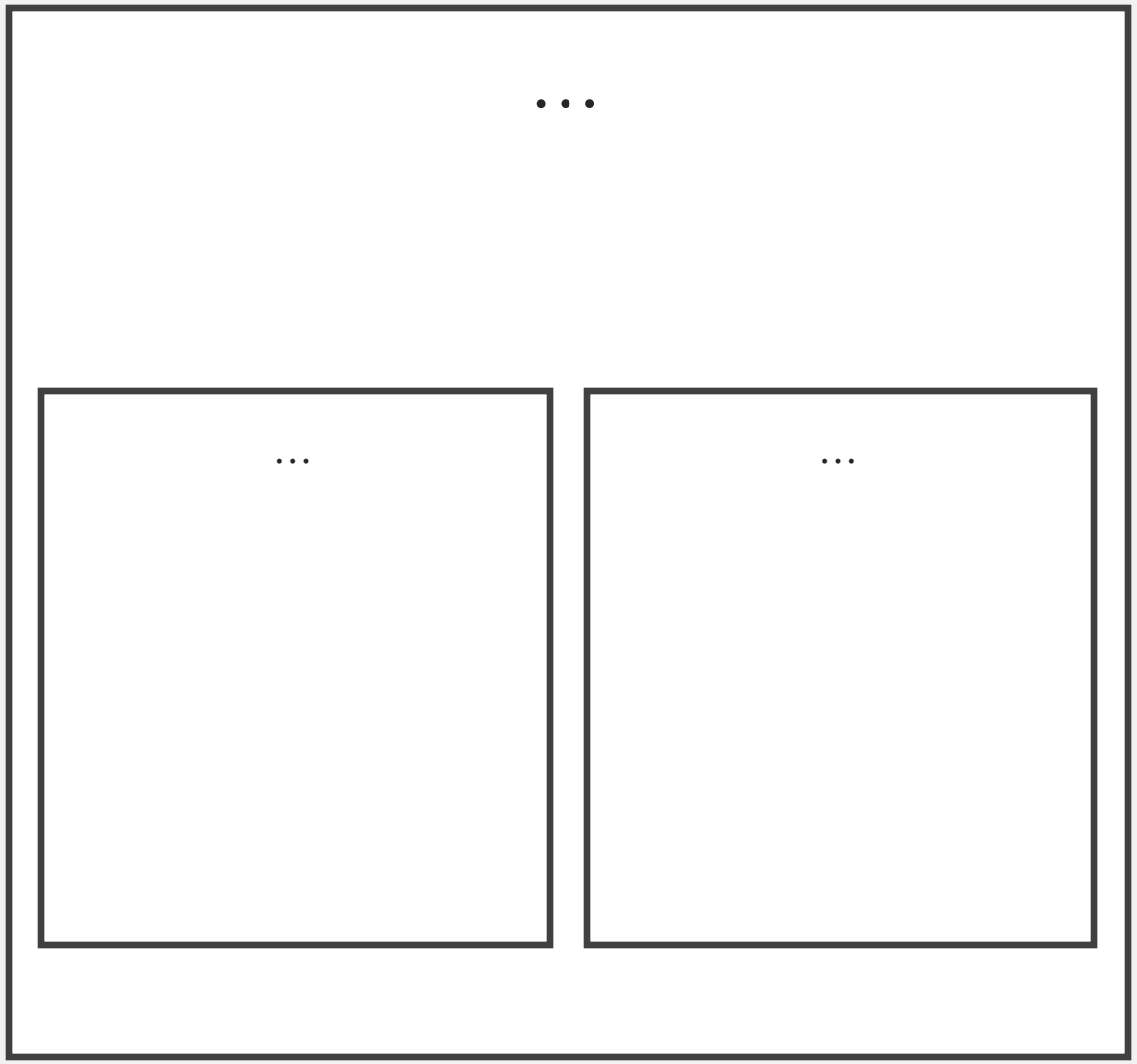
- Ensure the data meets the information need



DIMENSIONS OF DATA QUALITY

Timeliness

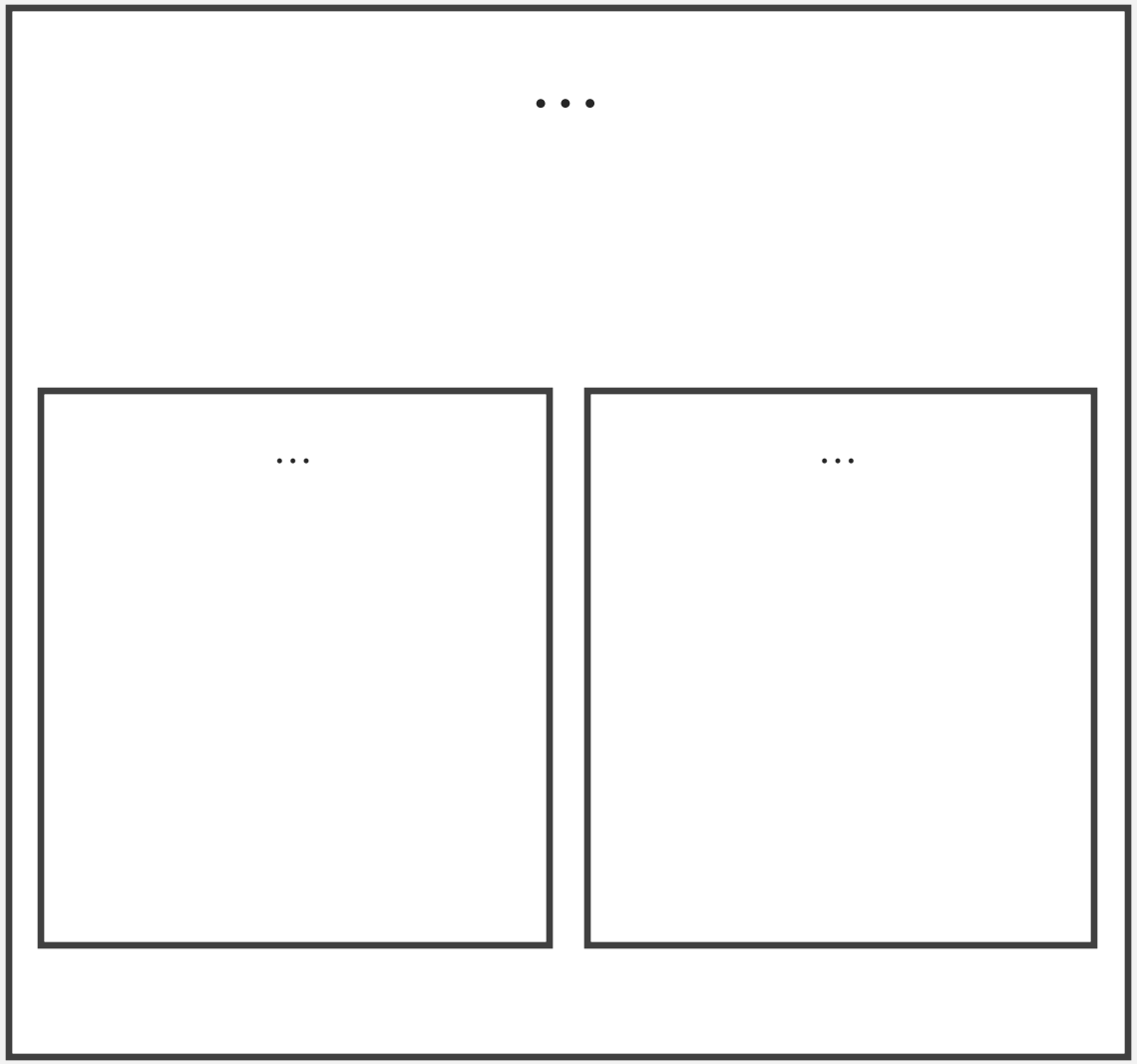
- Time to generate data output for use



DIMENSIONS OF DATA QUALITY

Uniqueness

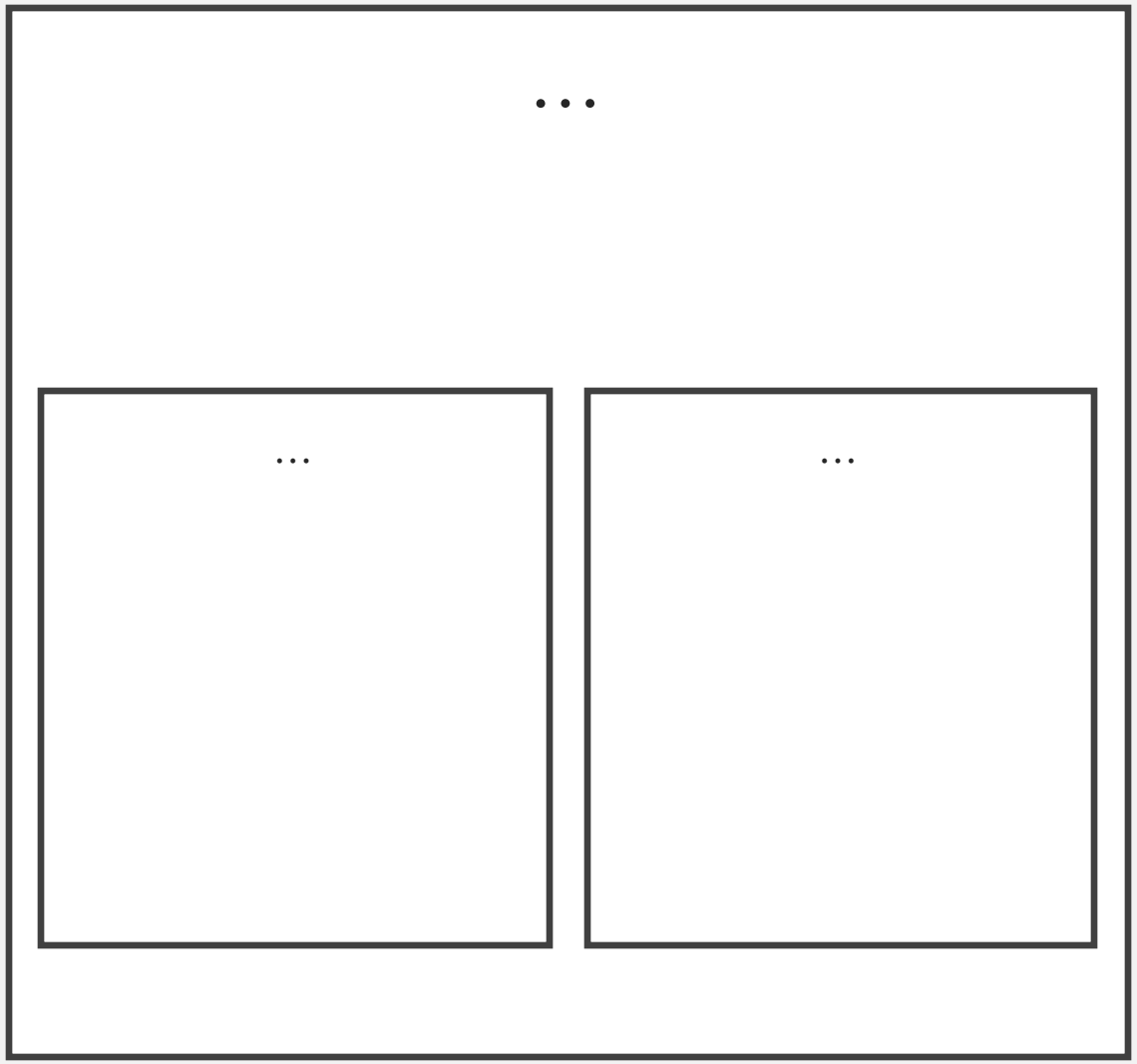
- Prevalence of duplicates



DIMENSIONS OF DATA QUALITY

Validity

- Format of data matches use requirements



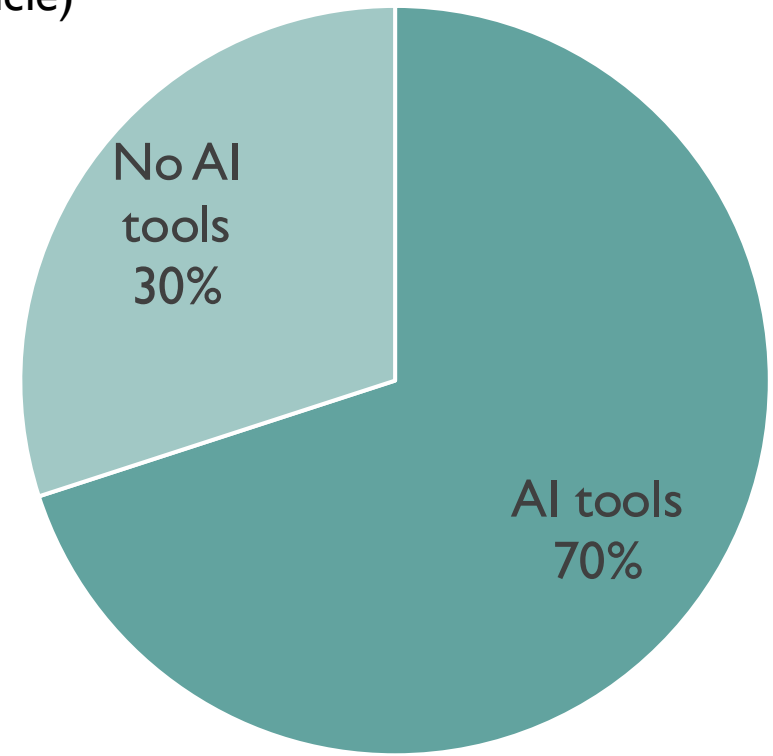
CASE STUDY

THE ARTICLE TO BE ANALYSED, THE ASSERTIONS MADE, AND THE ASSOCIATED SOURCES.

Assertion

- 70% of developers are already using or planning to use AI tools.

Source: Stack Overflow Developer Survey 2023 (cited in Werner Vogels' article)



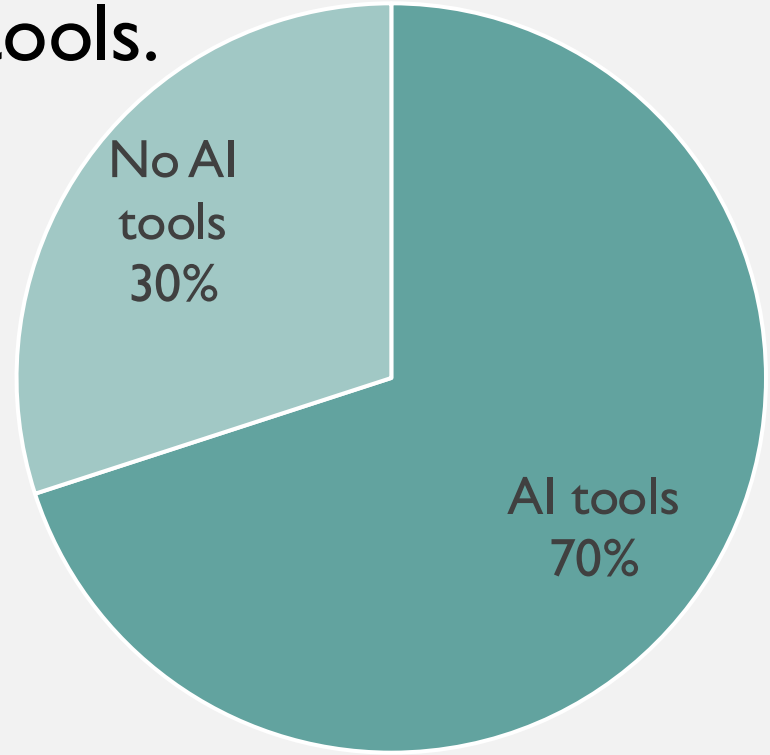
Data Source

ACTIVITY 2: DIMENSIONS SCAN

Assertion

- 70% of developers are already using or planning to use AI tools.

Source: Stack Overflow Developer Survey 2023 (cited in Werner Vogels' article)



- Accuracy
- Completeness
- Consistency
- Fitness for purpose
- Timeliness
- Uniqueness
- Validity

ACTIVITY 2: CRITICAL EVALUATION

Discuss the one of the following questions in your groups based on the Dimensions of Data Quality.

ACTIVITY 2: CRITICAL EVALUATION

1. Accuracy	Does the 70% figure reflect verifiable, real-world usage, or is it just subjective self-reporting from survey participants?
2. Completeness	Does a community survey (Stack Overflow) represent the entire engineering industry? Who is missing (e.g., highly regulated or legacy sectors)?
3. Consistency	Does this AI adoption rate align with findings from other reputable industry reports (e.g., Gartner or Microsoft) from the same timeframe?
4. Fitness for purpose	Is a general community survey the right "tool" to support high-level assertions about professional release lifecycles and product strategy?
5. Timeliness	Is 2023 survey data a high-quality "source of truth" for AI trends in 2026, given the exponential growth of Generative AI?
6. Uniqueness	How does the source ensure it represents unique individuals rather than data skewed by duplicate responses from the same users?
7. Validity	Does "planning to use" equate to "real productivity impact"? Is this a valid measure for judging actual technical advancements?

ACTIVITY 2: RE-EVALUATING THE ASSERTION

Re-evaluating the Assertion

(1 = Not at all credible, 10 = Highly credible)

STUDENT FEEDBACK – 2026 SEMESTER 1

Presentation Stream: **A**

Presentation Subtopic: Data Quality

URL: https://go.unimelb.edu.au/hoe2